

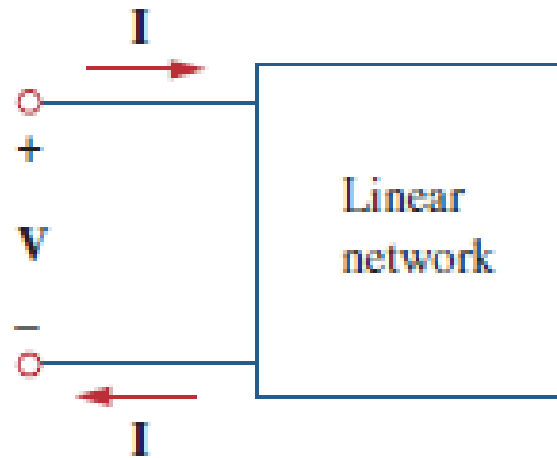
Module - 4

Two – Port Networks

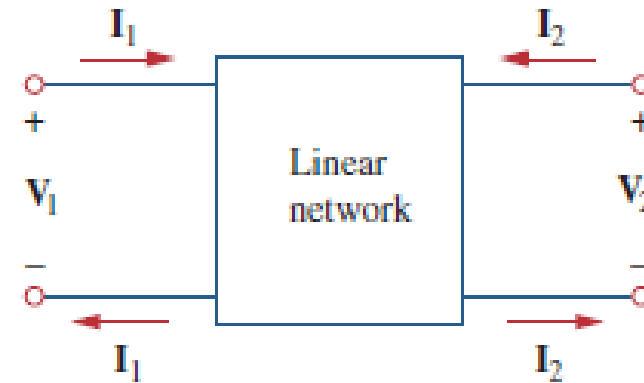
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SENSE

Introduction

- A pair of terminals through which a current may enter or leave a network is known as a port.
- Two terminal devices or elements (R, C and L) result in one port networks.



One-port network,

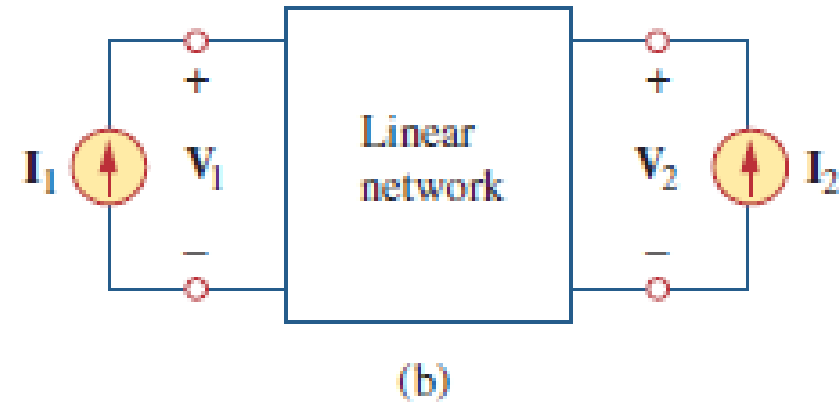
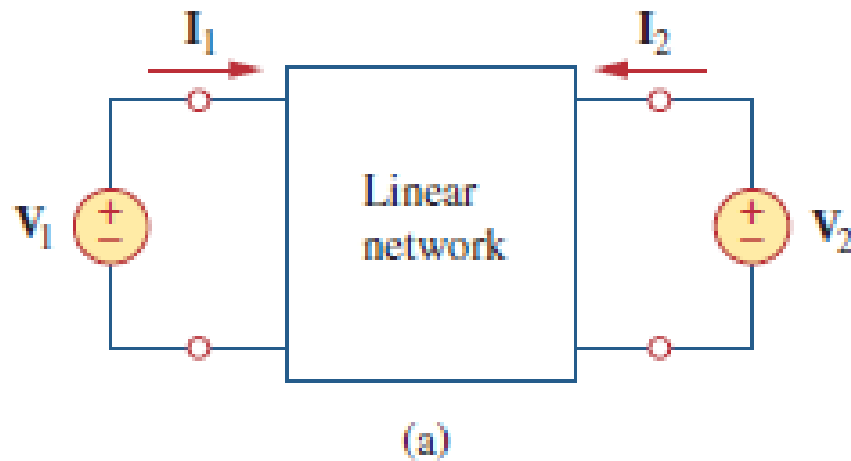


two-port networks

- In general, a network may have n ports.
- A port is an access to the network and consists of a pair of terminals; the current entering one terminal leaves through the other terminal so that the net current entering the ports equals zero.
- A two – port network is an electrical network with two separate ports for input and output.
- To characterize a two port network requires the relation between terminal quantities V_1 , V_2 , I_1 and I_2 , out of which two are independent.
- The various terms that relate these voltages and currents are called parameters.

Impedance Parameters

- Impedance and admittance parameters are commonly used in the synthesis of filters.
- They are useful in the design and analysis of impedance-matching networks and power distribution networks.



The linear two-port network: (a) driven by voltage sources, (b) driven by current sources.

- The terminal voltages can be related to the terminal currents as

$$\begin{aligned} \mathbf{V}_1 &= \mathbf{z}_{11}\mathbf{I}_1 + \mathbf{z}_{12}\mathbf{I}_2 \\ \mathbf{V}_2 &= \mathbf{z}_{21}\mathbf{I}_1 + \mathbf{z}_{22}\mathbf{I}_2 \end{aligned}$$

- Matrix form:

$$\begin{bmatrix} \mathbf{V}_1 \\ \mathbf{V}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{z}_{11} & \mathbf{z}_{12} \\ \mathbf{z}_{21} & \mathbf{z}_{22} \end{bmatrix} \begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix} = [\mathbf{z}] \begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix}$$

- Where z terms are called impedance parameters or simply z parameters or open circuit parameters with the units of ohms.

The values of the parameters can be evaluated by setting $I_1 = 0$ (input port open-circuited) or $I_2 = 0$ (output port open-circuited). Thus,

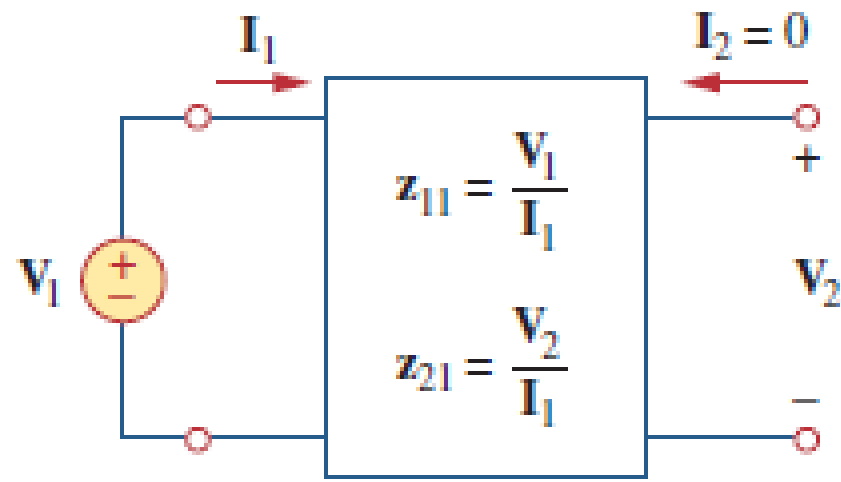
$$\begin{aligned} z_{11} &= \left. \frac{V_1}{I_1} \right|_{I_2=0}, & z_{12} &= \left. \frac{V_1}{I_2} \right|_{I_1=0} \\ z_{21} &= \left. \frac{V_2}{I_1} \right|_{I_2=0}, & z_{22} &= \left. \frac{V_2}{I_2} \right|_{I_1=0} \end{aligned}$$

z_{11} = Open-circuit input impedance

z_{12} = Open-circuit transfer impedance from port 1 to port 2

z_{21} = Open-circuit transfer impedance from port 2 to port 1

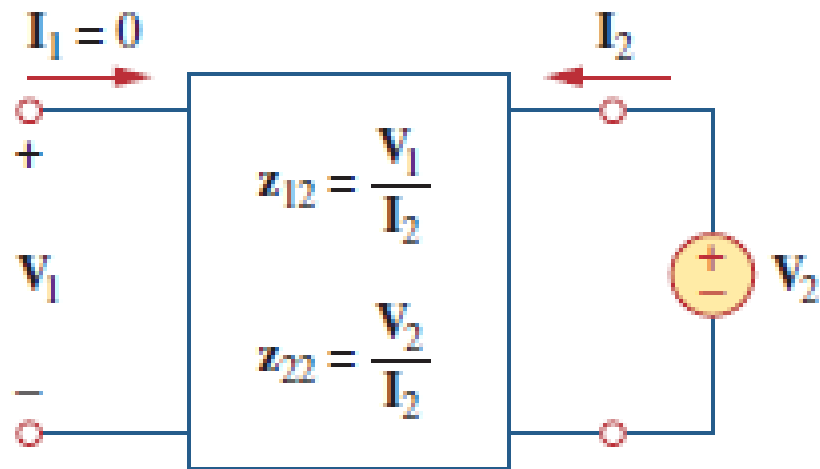
z_{22} = Open-circuit output impedance



(a)

Determination of the z parameters:
 (a) finding z_{11} and z_{21} , (b) finding z_{12} and z_{22} .

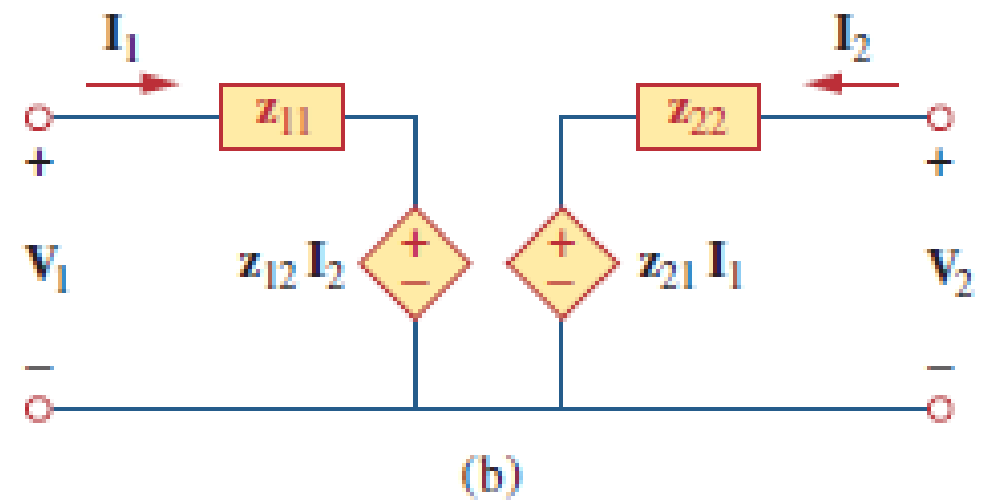
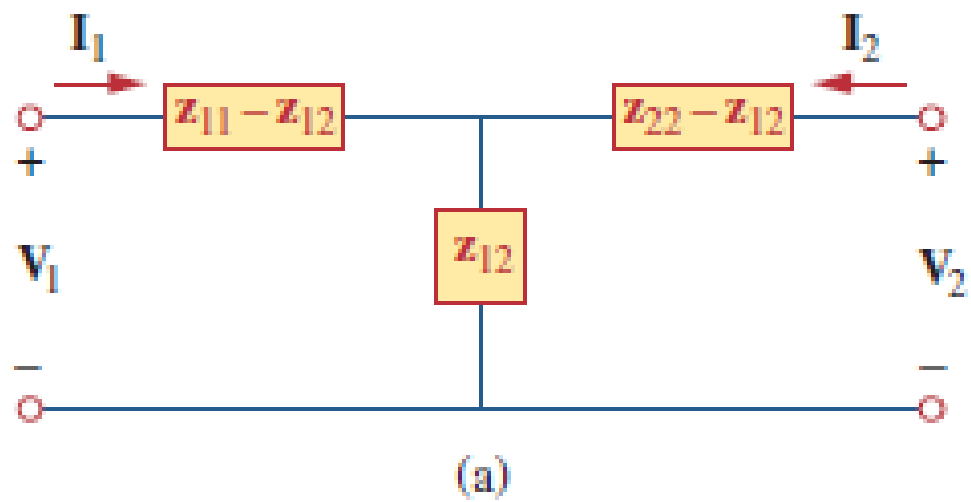
$$z_{11} = \frac{V_1}{I_1}, \quad z_{21} = \frac{V_2}{I_1}$$



(b)

$$z_{12} = \frac{V_1}{I_2}, \quad z_{22} = \frac{V_2}{I_2}$$

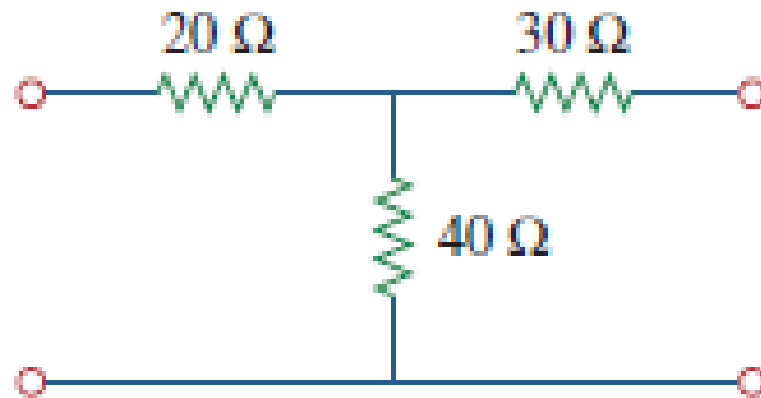
- Z_{11} and z_{22} are called as driving point impedances
- Z_{12} and z_{21} are called as transfer impedances
- $Z_{11} = Z_{22}$, the two port network is said to be symmetrical
- When the two-port network is linear and has no dependent sources, the transfer impedances are equal ($Z_{12} = Z_{21}$), and the two-port is said to be *reciprocal*.
- A reciprocal network can be replaced by the T-equivalent circuit.
- If the network is not reciprocal, a more general equivalent network is used.



(a) T-equivalent circuit (for reciprocal case only), (b) general equivalent circuit.

Numerical - 1

Determine the z parameters for the circuit



Solution:

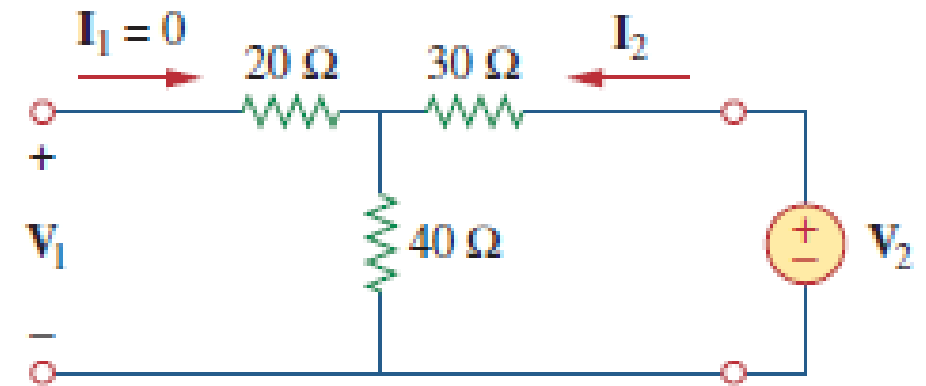
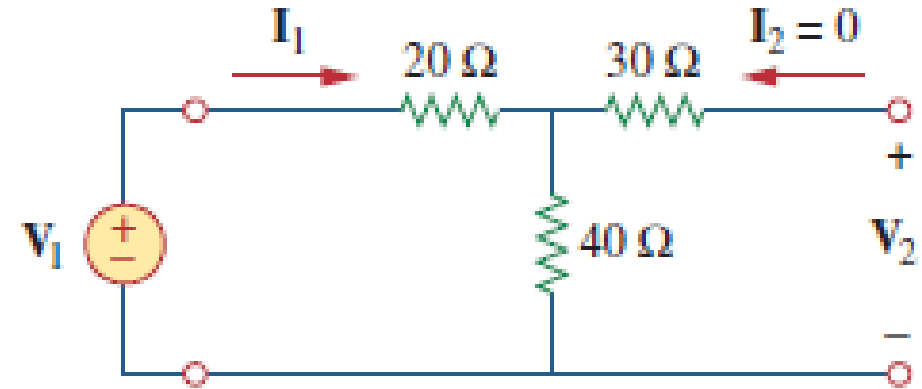
■ **METHOD 1**

$$z_{11} = \frac{V_1}{I_1} = \frac{(20 + 40)I_1}{I_1} = 60 \, \Omega$$

$$z_{21} = \frac{V_2}{I_1} = \frac{40I_1}{I_1} = 40 \, \Omega$$

$$z_{12} = \frac{V_1}{I_2} = \frac{40I_2}{I_2} = 40 \, \Omega,$$

$$z_{22} = \frac{V_2}{I_2} = \frac{(30 + 40)I_2}{I_2} = 70 \, \Omega$$



$$[z] = \begin{bmatrix} 60 \, \Omega & 40 \, \Omega \\ 40 \, \Omega & 70 \, \Omega \end{bmatrix}$$

■ **METHOD 2** Alternatively, since there is no dependent source in the given circuit, $z_{12} = z_{21}$

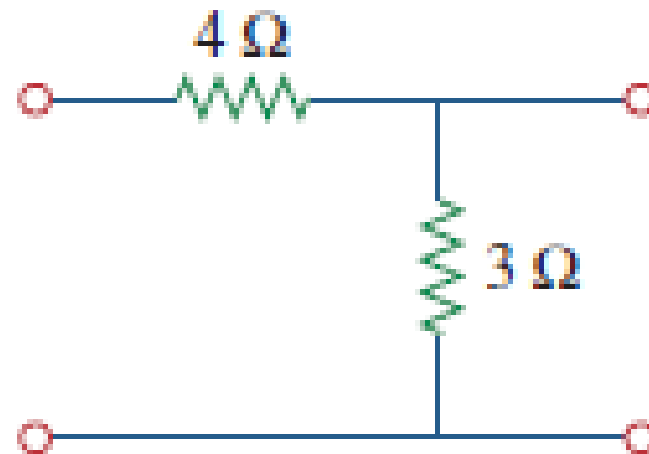
$$z_{12} = 40 \, \Omega = z_{21}$$

$$z_{11} - z_{12} = 20 \quad \Rightarrow \quad z_{11} = 20 + z_{12} = 60 \, \Omega$$

$$z_{22} - z_{12} = 30 \quad \Rightarrow \quad z_{22} = 30 + z_{12} = 70 \, \Omega$$

Numerical - 2

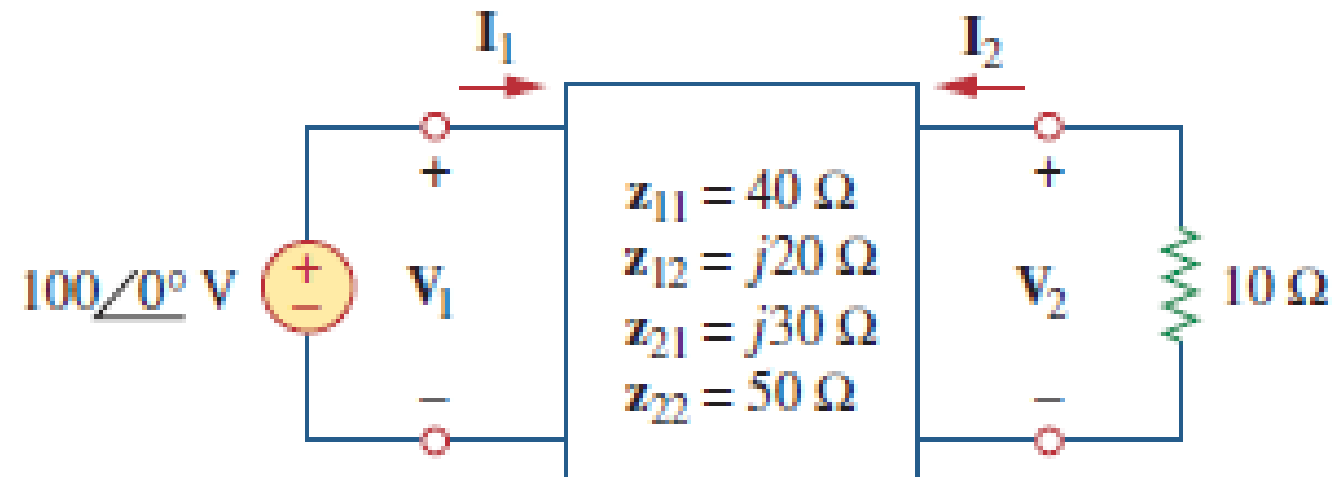
Find the z parameters of the two-port network



Answer: $z_{11} = 7\ \Omega$, $z_{12} = z_{21} = z_{22} = 3\ \Omega$.

Numerical - 3

Find I_1 and I_2 in the circuit



Solution:

This is not a reciprocal network.

$$\mathbf{V}_1 = 40\mathbf{I}_1 + j20\mathbf{I}_2$$

$$\mathbf{V}_2 = j30\mathbf{I}_1 + 50\mathbf{I}_2$$

Since we are looking for $\mathbf{I}_1$ and $\mathbf{I}_2$, we substitute

$$\mathbf{V}_1 = 100\angle 0^\circ, \quad \mathbf{V}_2 = -10\mathbf{I}_2$$

$$100 = 40\mathbf{I}_1 + j20\mathbf{I}_2$$

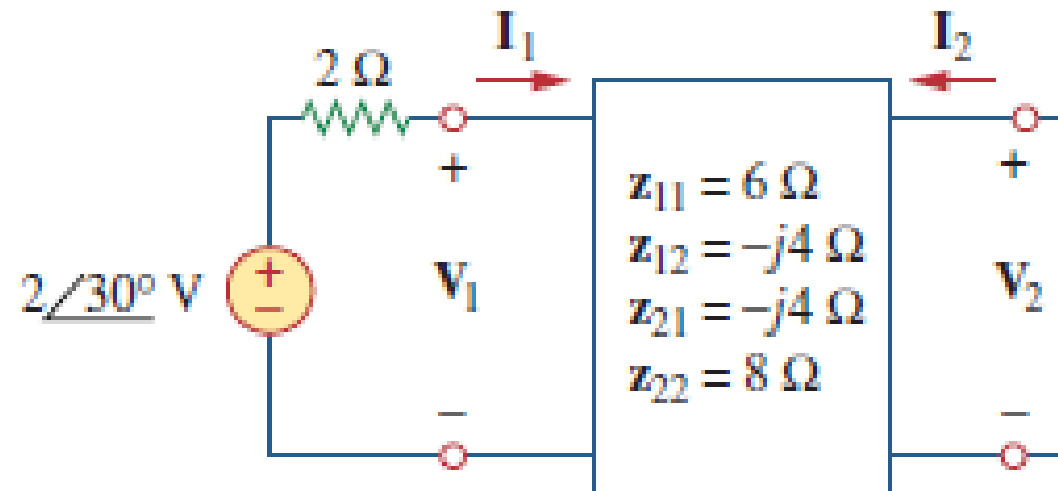
$$-10\mathbf{I}_2 = j30\mathbf{I}_1 + 50\mathbf{I}_2 \quad \Rightarrow \quad \mathbf{I}_1 = j2\mathbf{I}_2$$

$$100 = j80\mathbf{I}_2 + j20\mathbf{I}_2 \quad \Rightarrow \quad \mathbf{I}_2 = \frac{100}{j100} = -j$$

$$\mathbf{I}_1 = 2\angle 0^\circ \text{ A}, \quad \mathbf{I}_2 = 1\angle -90^\circ \text{ A}$$

Numerical-4

Calculate I_1 and I_2 in the two-port



Answer: $200\angle 30^\circ \text{ mA}$, $100\angle 120^\circ \text{ mA}$.